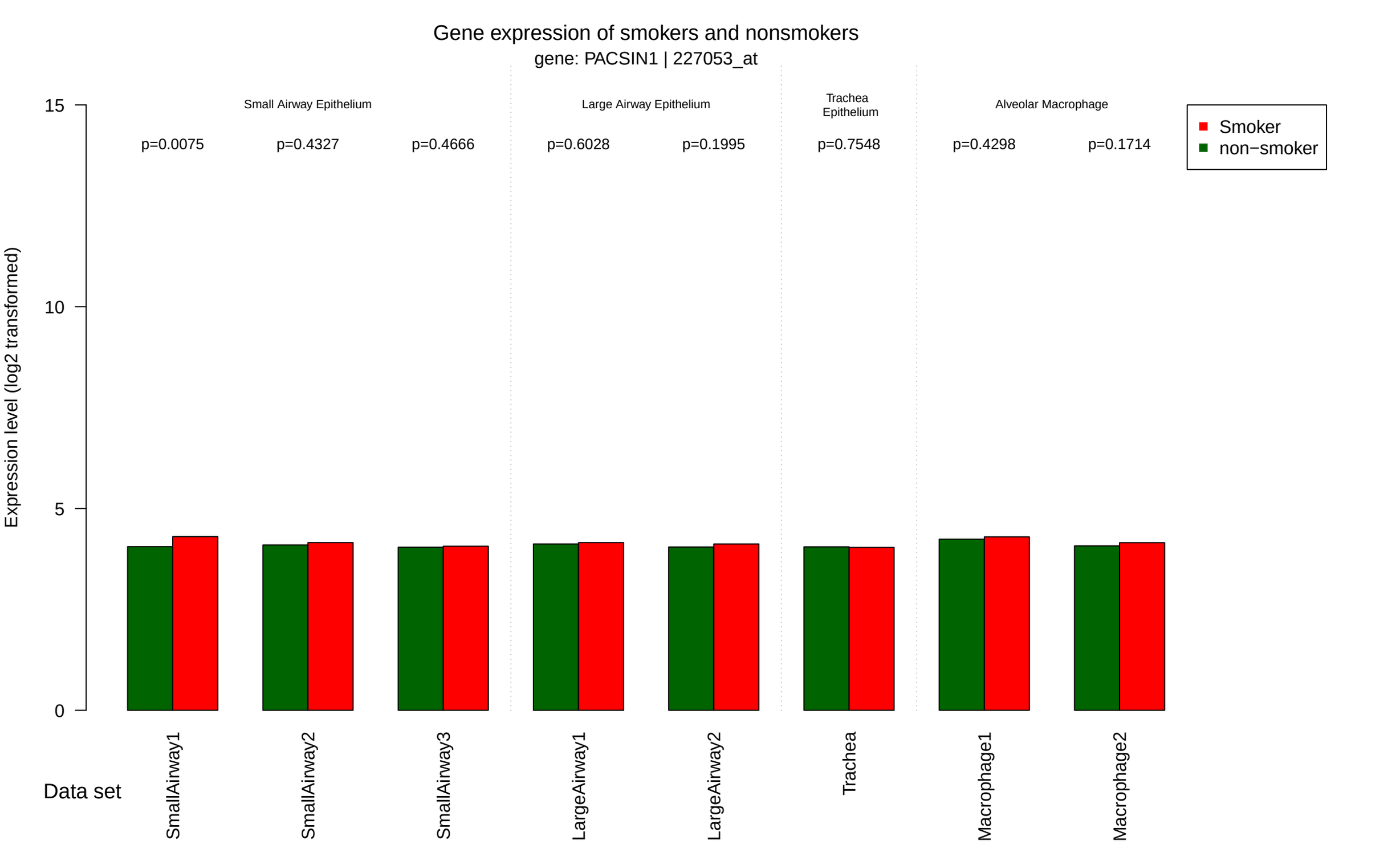
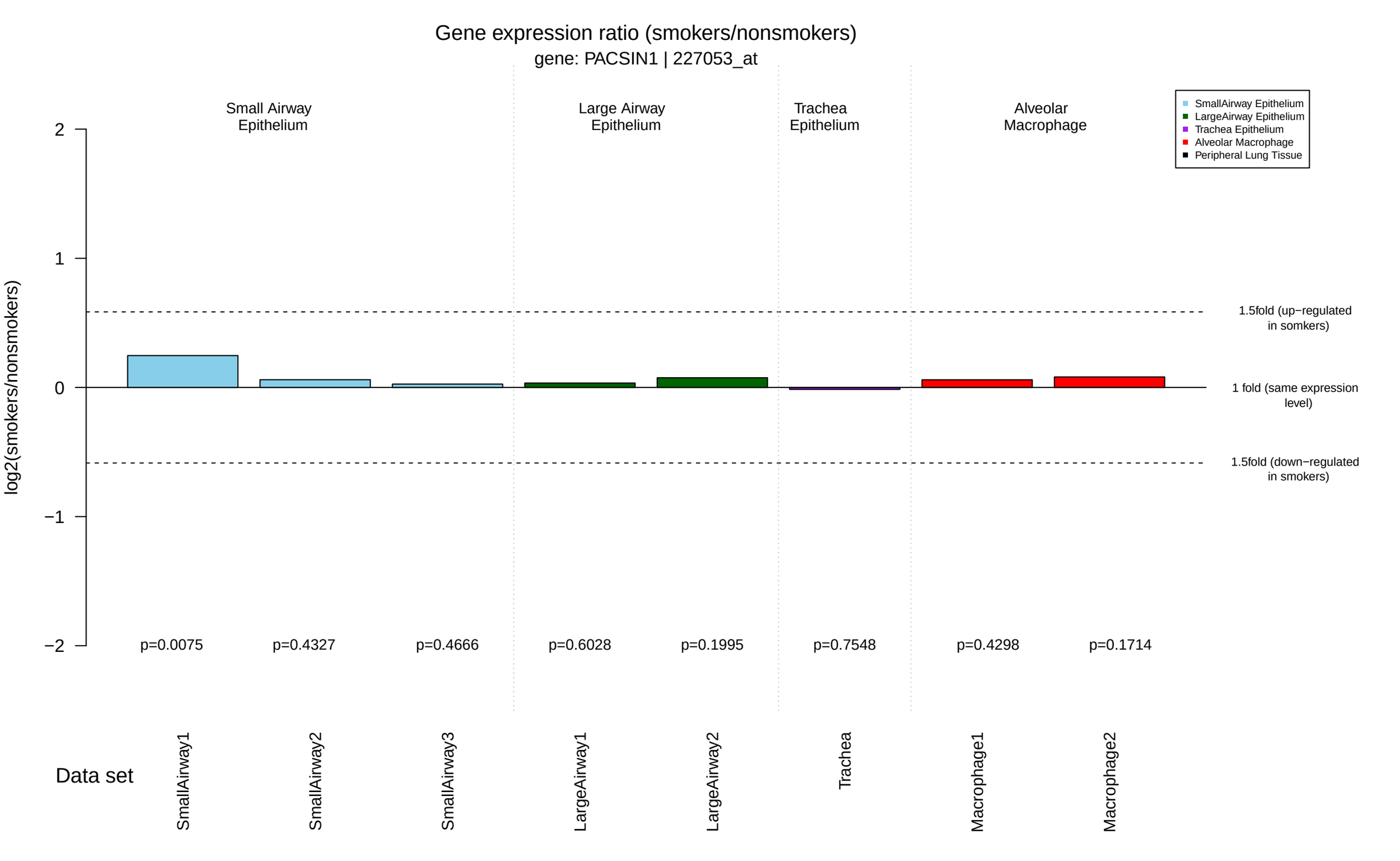
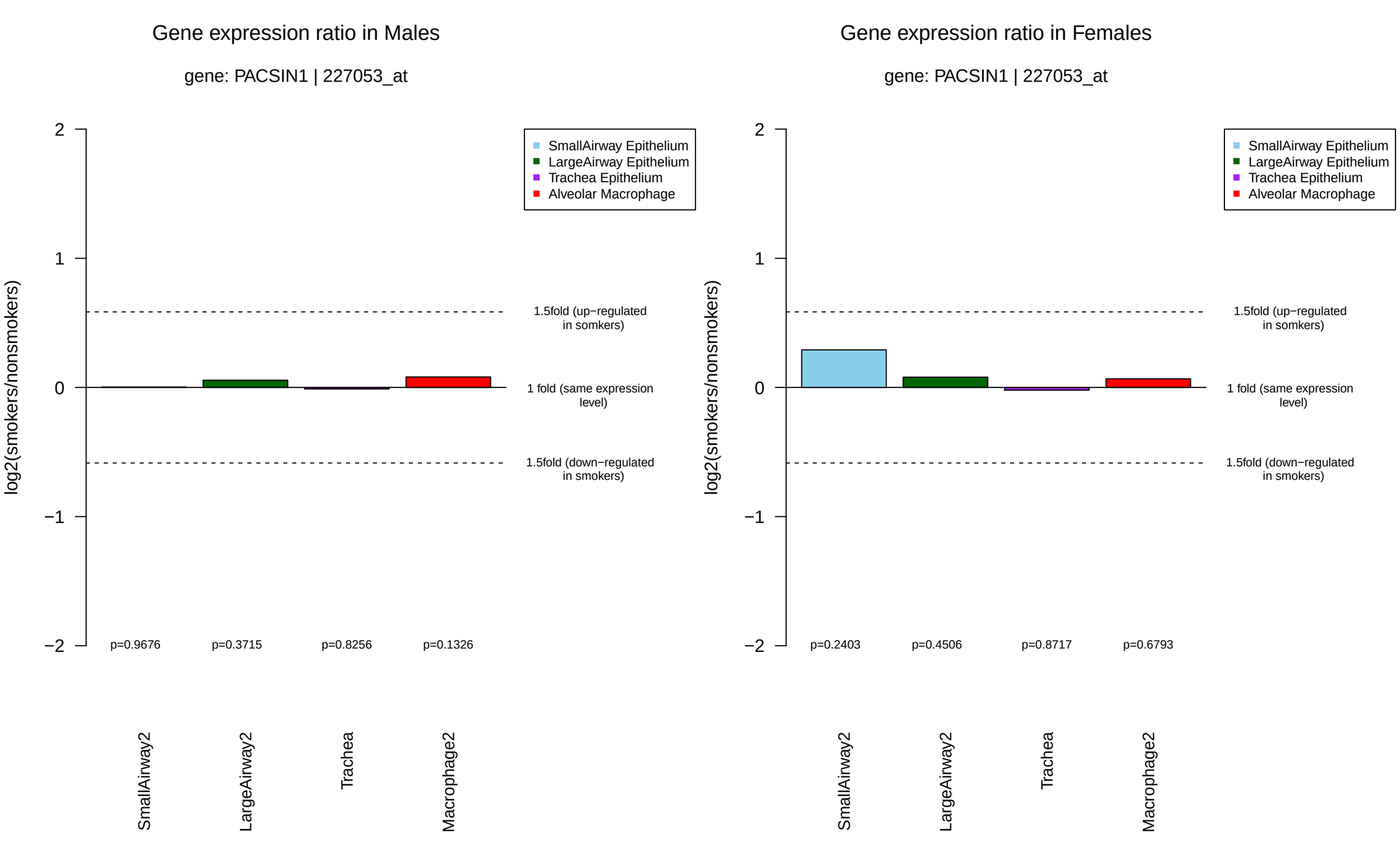
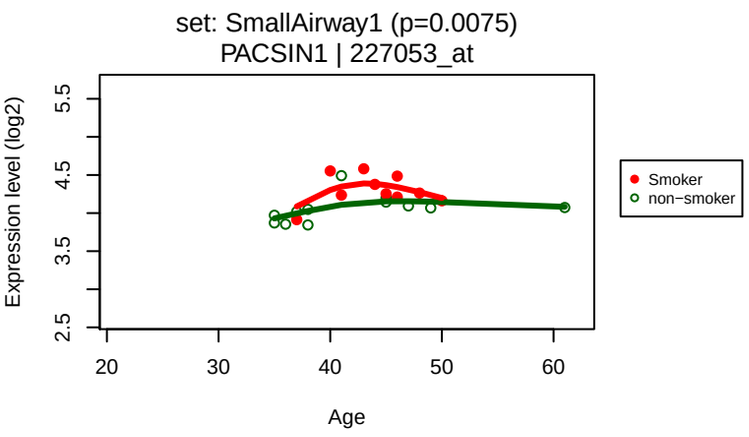
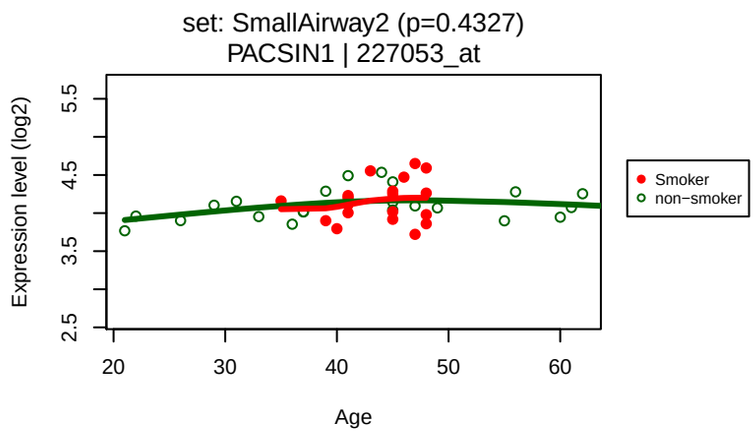
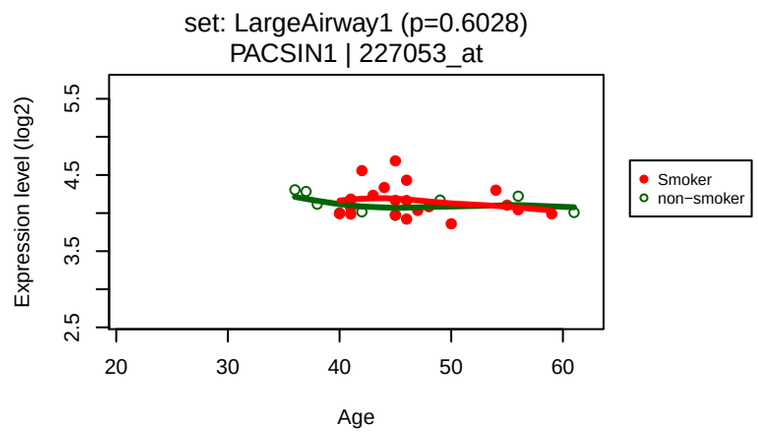

Gene expression of smokers and nonsmokers
gene: PACSIN1 | 227053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0075
p=0.4327
p=0.4666
p=0.6028
p=0.1995
p=0.7548
p=0.4298
p=0.1714
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

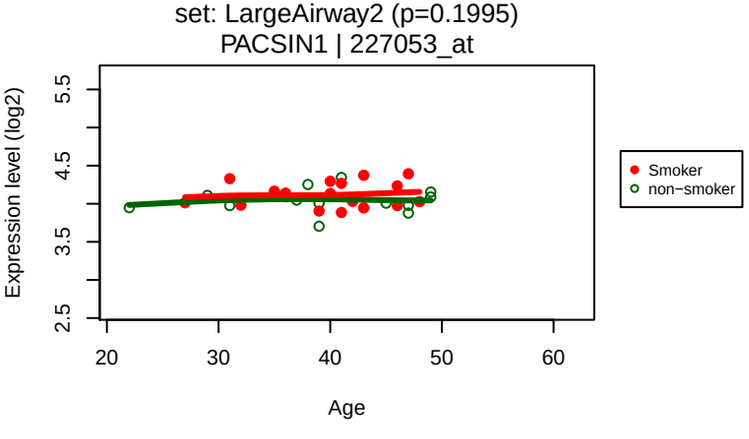

Gene expression ratio (smokers/nonsmokers)
gene: PACSIN1 | 227053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0075
p=0.4327
p=0.4666
p=0.6028
p=0.1995
p=0.7548
p=0.4298
p=0.1714
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

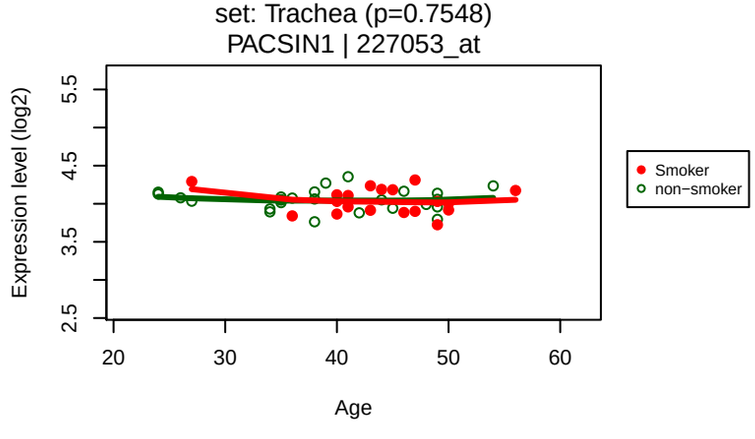

Gene expression ratio in Males
gene: PACSIN1 | 227053_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9676
p=0.3715
p=0.8256
p=0.1326
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PACSIN1 | 227053_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2403
p=0.4506
p=0.8717
p=0.6793
SmallAirway2
LargeAirway2
Trachea
Macrophage2

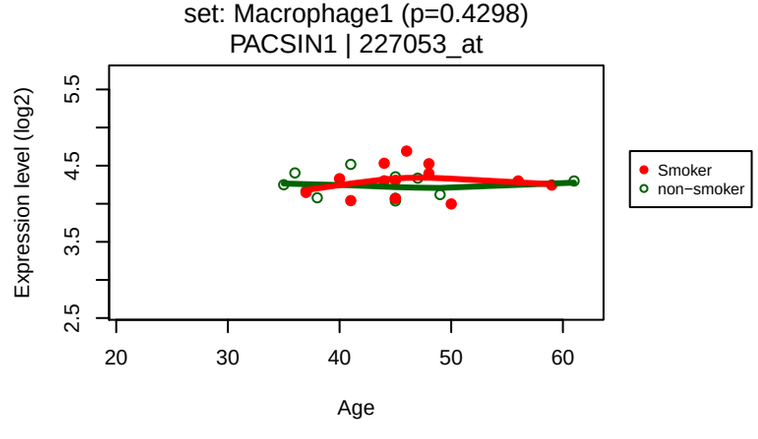

set: SmallAirway1 (p=0.0075)
PACSIN1 | 227053_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

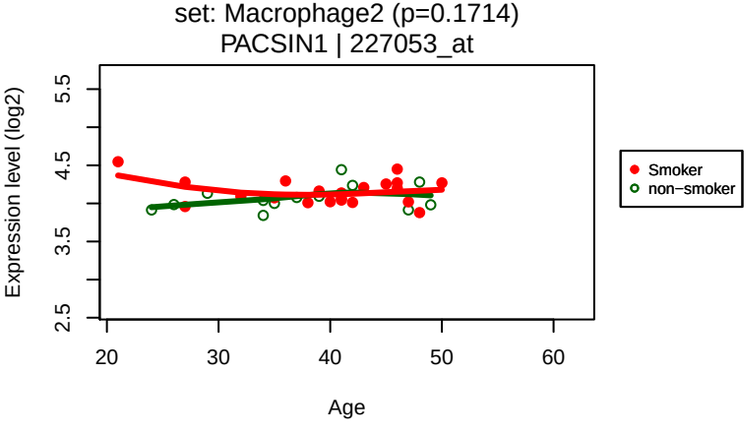

set: SmallAirway2 (p=0.4327)
PACSIN1 | 227053_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

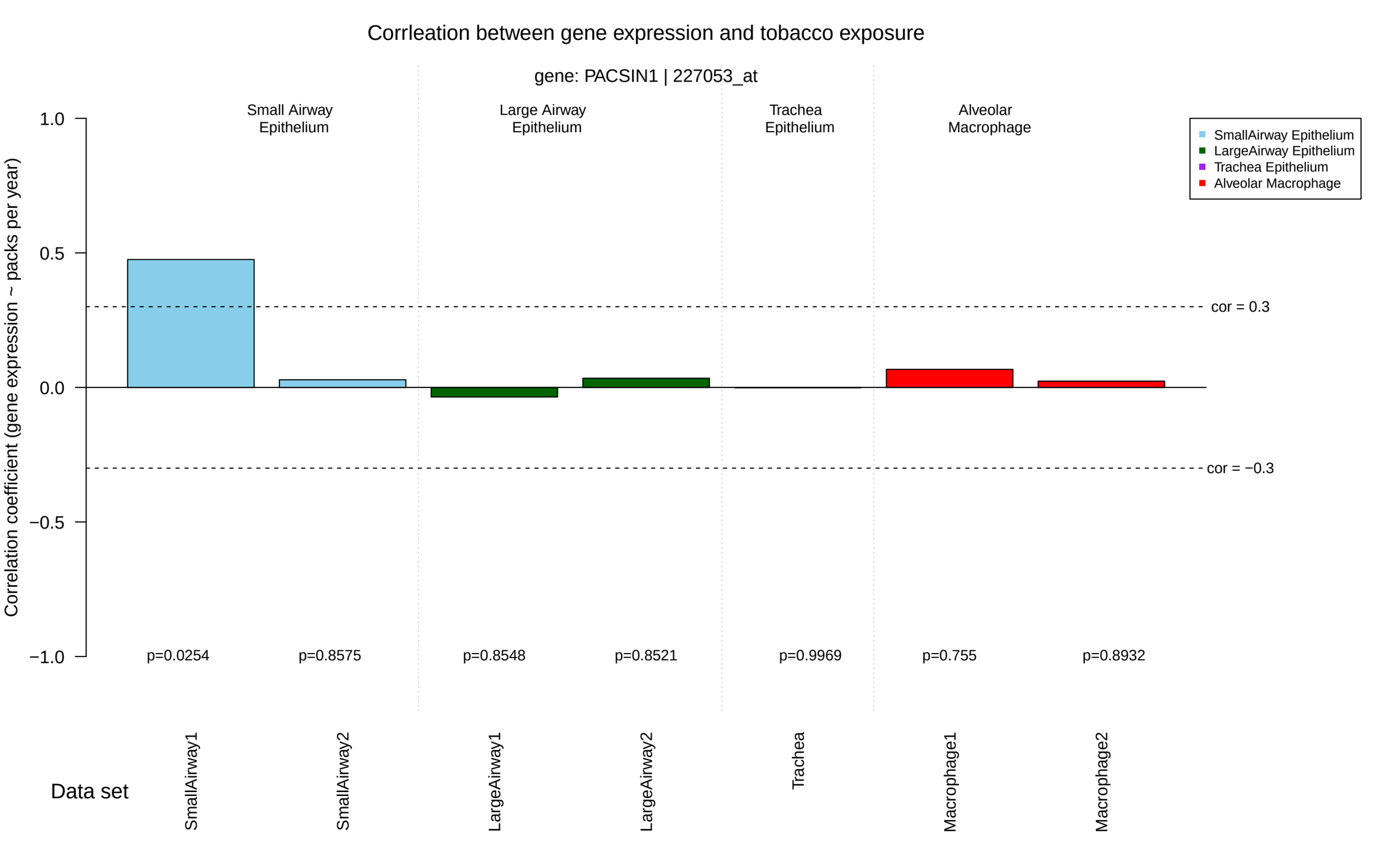

set: LargeAirway1 (p=0.6028)
PACSIN1 | 227053_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1995)
PACSIN1 | 227053_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7548)
PACSIN1 | 227053_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4298)
PACSIN1 | 227053_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1714)
PACSIN1 | 227053_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PACSIN1 | 227053_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0254
p=0.8575
p=0.8548
p=0.8521
p=0.9969
p=0.755
p=0.8932
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2